# Guerlain Lambert, PhD student in mathematics

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## Education

2023 – 2026 **PhD**, Centrale Lyon, Institut Camille Jordan.
Thesis title: *Sequential approach for meta-modeling and sensitivity analysis of computational codes in the presence of dependent random variables. Application to a design tool for vegetated strips.*
Supervisors: Céline Helbert (Centrale Lyon) and Claire Lauvernet (INRAE).

2021 – 2023 **Master in Applied Mathematics**, Maths en Actions, Université Claude Bernard Lyon 1.
Statistics, Machine Learning and application to environment, climate change.
Internship: *Approach for Meta-Modeling and Sensitivity Analysis of Computational Codes in the Presence of Dependent Random Variables. Application to a design tool for vegetated strips*, supervised by Céline Helbert (Centrale Lyon) and Claire Lauvernet (INRAE).

2020 – 2021 **Bachelor of mathematics**, Université Paul Sabatier Toulouse 3.

2017 – 2019 **CPGE**, Lycée Victor Hugo, Caen.

2017 **Baccalauréat**, Lycée Alexis de Tocqueville, Cherbourg-en-Cotentin.

## Research Publications

### Preprints

1. G. Lambert, C. Helbert, and C. Lauvernet. "Gradient-based active learning with gaussian processes for global sensitivity analysis." (2026).

### Accepted Articles

1. G. Lambert, C. Helbert, and C. Lauvernet, "Quantization-based latin hypercube sampling for dependent inputs with an application to sensitivity analysis of environmental models," *Applied Stochastic Models in Business and Industry*, vol. 41, no. 3, e2899, 2025. DOI: 10.1002/asmb.2899.

## Conferences, seminars, summer schools, workshops...

2026 **RT-UQ**, ENSAI, Rennes (France), talk (Best Oral presentation Award).
Gradient-Based Active Learning with Gaussian Processes for Global Sensitivity Analysis.

2025 **ENBIS**, University of Piraeus (Greece), talk.
Gradient-Based Active Learning with Gaussian Processes for Global Sensitivity Analysis.

**Journées de la statistique**, University of Aix-Marseille (France), talk.
Gradient-Based Active Learning with Gaussian Processes for Global Sensitivity Analysis.

**SAMO RT-UQ**, University of Grenoble (France), poster (Best Poster Award).
Surrogate-based active learning for Sobol' indices estimation with dependent inputs.

2024 **CIROQUO**, Ecole des Mines de Saint-Étienne (France), poster.
Quantization-based LHS for dependent inputs : application to sensitivity analysis of environmental models.

## Conferences, seminars, summer schools, workshops... (continued)

- **ETICS**, Saissac (France), talk.
  Quantization-based LHS for dependent inputs : application to sensitivity analysis of environmental models.
- **CIROQUO**, CEA-DAM, Bruyère-le-Châtel (France), poster.
  Quantization-based LHS for dependent inputs : application to sensitivity analysis of environmental models.
- **MEXICO**, INRAE Lyon (France), talk.
  Quantization-based LHS for dependent inputs : application to sensitivity analysis of environmental models.

2023
- **CIROQUO**, INRIA Saclay (France), poster.
  Deep Gaussian Process Metamodeling with Mixed Variables for Vegetative Filter Strip Design in Agricultural Context.
- **MEXICO**, AgroParisTech - INRAE, Saclay (France), talk.
  Approach for Meta-Modeling and Sensitivity Analysis of Computational Codes in the Presence of Dependent Random Variables.

## Awards

2026
- **Best Oral presentation Award**, RT-UQ / MASCOT-NUM 2026, Rennes (France).

2025
- **Best Poster Award**, RT-UQ / MASCOT-NUM 2025, Grenoble (France).

## Teaching Experience

174h in total at Centrale Lyon.

2026
- **Machine Learning and Applications tutorials (8h)**, M1, Centrale Lyon.

2025
- **Probability-Statistics tutorials (34h)**, L3, Centrale Lyon.
- **Statistics for engineers tutorials (12h)**, M2, Centrale Lyon.

2024
- **Probability-Statistics tutorials (40h)**, L3, Centrale Lyon.
- **Statistics for engineers tutorials (16h)**, M2, Centrale Lyon.
- **Machine Learning and Applications tutorials (8h)**, M1, Centrale Lyon.

2023
- **Probability-Statistics tutorials (16h)**, L3, Centrale Lyon.
- **Numerical analysis tutorials (30h)**, L3, Centrale Lyon.
- **Applied Analysis tutorials (4h)**, L3, Centrale Lyon.
- **Machine Learning and Applications tutorials (6h)**, M1, Centrale Lyon.

## Skills

Languages
- French (native language) and English (C1 level)

Coding
- Python (PyTorch, GPyTorch, BoTorch, sklearn), R, Matlab, LaTeX.